

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO2.2a	B1	$\begin{bmatrix} 5 \\ -1 \\ 3 \end{bmatrix}$
Total 1 mark			

Q2.

(a) $\lambda = -1$

$\lambda = -1$ seen or implied

B1

$\lambda = -1$ verified in all three components

Shown

B1

2

(b) $\pm \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$

$\overline{AB}$ or $\overline{BA}$ correct

B1

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$$\mathbf{r} = \overline{OA} + \mu \overline{AB} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + \mu \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$$

$\mathbf{a} + \mu \mathbf{d}$

M1

OE; ft on $\overline{AB}$ or $\overline{BA}$

A1ft

3

(c) $\overline{CD} = \overline{OD} - \overline{OC}$

$$= \begin{bmatrix} 3-2\mu \\ -2-3\mu \\ 4+2\mu \end{bmatrix} - \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 7-2\mu \\ -7-3\mu \\ 5+2\mu \end{bmatrix}$$

$\pm \overline{CD}$ in terms of μ

OE

B1

$$\overline{CD} \cdot \overline{AB} = 0 \quad \text{or} \quad \overline{CD} \cdot \overline{AD} = 0$$

$$= \left(\begin{bmatrix} 3-2\mu \\ -2-3\mu \\ 4+2\mu \end{bmatrix} - \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix} \right) \cdot \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix} = 0$$

Candidate's $\overline{CD}$ sp with candidate's $\overline{AB}$ or $\overline{AD} = 0$
PI by a solution for μ

M1

$$-14 + 4\mu + 21 + 9\mu + 10 + 4\mu = 0$$

$$17 + 17\mu = 0$$

$$\mu = -1$$

Expand sp to an equation in μ and solve for μ

m1A1

D is at (5, 1, 2)

Accept as a column vector

A1

5

Alternative using Pythagoras

$$\overline{CD} = \overline{OD} - \mu \overline{OC}$$

$$= \begin{bmatrix} 3-2\mu \\ -2-3\mu \\ 4+2\mu \end{bmatrix} - \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 7-2\mu \\ -7-3\mu \\ 5+2\mu \end{bmatrix}$$

$\pm \overline{CD}$ in terms of μ

(B1)

$$AC^2 = AD^2 + CD^2$$

$$(7^2 + 7^2 + 5^2) = \mu^2(2^2 + 3^2 + 2^2) + ((7-2\mu)^2 + (7+3\mu)^2 + (5+2\mu)^2)$$

$$\overline{AC} = \begin{bmatrix} -7 \\ 7 \\ -5 \end{bmatrix} \quad \overline{AD} = \begin{bmatrix} -2\mu \\ -3\mu \\ 2\mu \end{bmatrix}$$

Correct Pythagoras expression in terms of μ ;

(M1)

$$123 = 17\mu^2 + 123 + 34\mu + 17\mu^2$$

Multiply out and solve to find a value for μ

(m1)

$$0 = 34\mu^2 + 34\mu$$

$$\mu = -1 \quad (\mu = 0 \text{ is point } A)$$

$$\mu = -1$$

(A1)

D is at (5, 1, 2)

(A1)

(5)

(d) $\overrightarrow{OE} = \overrightarrow{OA} + \overrightarrow{AE} = \overrightarrow{OA} + 3\overrightarrow{AD}$

Accept $AE = 3AD$

M1

$$\overrightarrow{OE} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \quad E \text{ is at } (9, 7, -2)$$

Accept as a column vector

A1

OR

$$\overrightarrow{OE} = \overrightarrow{OA} + \overrightarrow{AE} = \overrightarrow{OA} + 3\overrightarrow{DA}$$

Accept $AE = 3DA$

M1

$$\overrightarrow{OE} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix} \quad E \text{ is at } (-3, -11, 10)$$

Accept as a column vector

A1

4

Alternative

$$|\overrightarrow{DE}| = 2|\overrightarrow{AD}| \Rightarrow \overrightarrow{OE} = \overrightarrow{OD} + 2\overrightarrow{AD}$$

(M1)

$$\overrightarrow{OE} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \quad E \text{ is at } (9, 7, -2)$$

(A1)

$$|\overrightarrow{DE}| = 4|\overrightarrow{DA}| \Rightarrow \overrightarrow{OE} = \overrightarrow{OD} + 4\overrightarrow{DA}$$

(M1)

$$\overrightarrow{OE} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} + 4 \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix} \quad E \text{ is at } (-3, -11, 10)$$

(A1)

(4)

[14]

Q3.

(a)

$$\left. \begin{array}{l} \text{direction ratios of line} = p : 3 : -1 \\ \text{normal to plane} = 1 : 1 : 2 \end{array} \right\} \begin{array}{l} \text{not} \\ \text{equal} \end{array}$$

Accept not parallel or showing vector product is non zero

B1

1

(b) $\begin{array}{l} x = 3 + pt \\ y = q + 3t \\ z = 1 - t \end{array}$

Parametric form seen

M1

$$\text{Meets plane} \Rightarrow 3 + pt + q + 3t + 2(1 - t) = 10$$

$$\Rightarrow (5 + q) + t(p + 1) = 10$$

Correct substitution in plane

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A1

$$\text{Within plane} \Rightarrow q = 5, p = -1$$

M1 – Finding one correct value

A1 – Both values correct

M1A1

4

Alternative:

Point (3, q, 1) is common to line and plane

$$\text{Hence } \begin{pmatrix} 3 \\ q \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10 \text{ which gives } q = 5$$

Uses common point to find q

(M1A1)

Another point common to both is $(3 + p, 8, 0)$

Hence $\begin{pmatrix} 3+p \\ 8 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10$ which gives $p = -1$

Use of second point and value of q to find p or

consideration of scalar product $\begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

(M1A1)

(4)

(c) (i) $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $\mathbf{d} = \begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix}$

Let α be angle between normal and direction ratios
(plane) (line)

$\mathbf{n} \cdot \mathbf{d} = p + 1$

$\mathbf{n} \cdot \mathbf{d}$ correct

M1

$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \cos \alpha = \frac{\pm 1}{\sqrt{6}}$

Correct $\cos \alpha$ stated or implied

B1

$\Rightarrow \frac{p+1}{\sqrt{6}\sqrt{p^2+10}} = \frac{\pm 1}{\sqrt{6}}$

*Forming equation connecting all relevant parts **and**
attempting to solve for p (condone missing $\pm$)
Dependent on first M1 – fully correct for A1*

m1A1

$\Rightarrow (p+1)^2 = p^2 + 10$

$\Rightarrow p^2 + 2p + 1 = p^2 + 10$

$\Rightarrow 2p = 9$ giving $p = 4.5$

CAO

A1

5

Alternative:



$$|\mathbf{n} \times \mathbf{d}| = \sqrt{49 + (1+2p)^2 + (3-p)^2}$$

$|\mathbf{n} \times \mathbf{d}|$ correct

(M1)

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \sin \alpha = \frac{\sqrt{5}}{\sqrt{6}}$$

Correct $\sin \alpha$ stated or implied

(B1)

$$\frac{\sqrt{49 + (1+2p)^2 + (3-p)^2}}{\sqrt{6}\sqrt{p^2+10}} = \frac{\sqrt{5}}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and** attempting to solve for p . Dependent on first M1 – fully correct for A1

(m1A1)

Leading to $p = 4.5$

CAO

(A1)

(5)

(ii) $z = 2 \Rightarrow t = -1 \Rightarrow x = -1.5$

$$p = 4.5 \quad y = q - 3$$

$$\Rightarrow -1.5 + q - 3 + 4 = 10$$

Attempt to form an equation for q using $t = -1$

M1

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$$q = 10.5$$

CAO

A1

2

[12]

Q4.

(a) $2\lambda = 8 + 2\mu$

$$-2 = 3 + 5\mu$$

$$\lambda = 3, \mu = -1$$

Use the first two equations to set up and attempt to solve simultaneous equations for λ or μ . Must not assume $q = 4$.

M1

$$q - \lambda = 5 + 4\mu$$

$$q = 5 + 3 - 4 = 4$$

Use 3rd equation to show $q = 4$ **AG**.

A1

p is at (6, -2, 1)

Condone as a column vector

B1

(b) $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} = 4 - 4 = 0 \Rightarrow \text{perpendicular}$
 or $2 \times 2 + -1 \times 4 = 0$ seen **and** conclusion
 (condone $\theta = 90$)

B1

1

(c) (i) A is at (2, -2, 3)
 $AP^2 = (6 - 2)^2 + (-2 - -2)^2 + (1 - 3)^2$

Candidate's $|AP|^2$

M1

$$= 20$$

CAO

$$NMS AP = \sqrt{20} \quad M1A0$$

A1

2

(ii) $(\overrightarrow{PB}) = \begin{bmatrix} 8 \\ 3 \\ 5 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} - \begin{bmatrix} 6 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + 2\mu \\ 5 + 5\mu \\ 4 + 4\mu \end{bmatrix}$

Clear attempt to find $\overrightarrow{BP}$ or $\overrightarrow{PB}$ in terms of μ

M1

$$(PB^2) = (2 + 2\mu)^2 + (5 + 5\mu)^2 + (4 + 4\mu)^2$$

Find distance BP in terms of μ

m1

$$45\mu^2 + 90\mu + 45 = 20$$

$$(5)(9\mu^2 + 18\mu + 5) = 0$$

Attempt to set up three-term quadratic in μ and equate to their AP^2

m1

$$(3\mu + 1)(3\mu + 5) = 0$$

Solve quadratic equation to obtain **two** values of μ

m1

$$\mu = -\frac{1}{3} \text{ and } \mu = -\frac{5}{3}$$

Both values correct.

A1

$$B \text{ is at } \left(\frac{22}{3}, \frac{4}{3}, \frac{11}{3}\right) \text{ and } \left(\frac{14}{3}, -\frac{16}{3}, -\frac{5}{3}\right)$$

Both sets of coordinates required.

Condone column vectors.

SC one value of μ correct and corresponding coordinates of B correct scores A1 A0.

A1

Alternative 1:

$$\overrightarrow{AB} = \begin{bmatrix} 8 \\ 3 \\ 5 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6+2\mu \\ 5+5\mu \\ 2+4\mu \end{bmatrix}$$

Clear attempt to find $\overrightarrow{AB}$ or $\overrightarrow{BA}$ in terms of μ

(M1)

$$(\overrightarrow{AB})^2 = (6+2\mu)^2 + (5+5\mu)^2 + (2+4\mu)^2$$

Find distance AB in terms of μ

(m1)

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$$45\mu^2 + 90\mu + 65 = 40$$

$$(5)(9\mu^2 + 18\mu + 5) = 0$$

Attempt to set up three-term quadratic in μ and equate to their $2 \times$ their AP^2

(m1)

As before

Alternative 2:

$$\overrightarrow{PB} = k \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$$

(M1)

$$k^2(2^2 + 5^2 + 4^2) = 20$$

$m1$ for LHS

(m1)

$m1$ for equating to 'their 20'
May score M1m0m1

(m1)

$$k = \pm \frac{2}{3}$$

(A1)

$$\overrightarrow{OB} = \overrightarrow{OP} + (\pm)(\text{their value of } k) \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$$

(m1)

$$B \text{ is at } \left(\frac{22}{3}, \frac{4}{3}, \frac{11}{3} \right) \text{ and } \left(\frac{14}{3}, -\frac{16}{3}, -\frac{5}{3} \right)$$

(A1)

6

[12]

Q5.

$$(a) \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -4 & 7 \\ 2 & -2 & 3 \end{vmatrix} = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

Attempt at the vector product of the 2 d.v.s **OR**
two scalar products = 0 and some manipulation attempt

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A1

$$(b) \quad (i) \quad \overrightarrow{AB} = \begin{bmatrix} 2\beta + 7 \\ 19 - 2\beta \\ 3\beta - 2 \end{bmatrix} - \begin{bmatrix} 3\alpha + 7 \\ -25 - 4\alpha \\ 7\alpha + 9 \end{bmatrix} = \begin{bmatrix} 2\beta - 3\alpha \\ 44 - 2\beta + 4\alpha \\ -11 + 3\beta - 7\alpha \end{bmatrix}$$

Good attempt

M1

$$\overrightarrow{AB} = \lambda \mathbf{n} = \begin{bmatrix} 2\lambda \\ 5\lambda \\ 2\lambda \end{bmatrix}$$

With their $\overrightarrow{AB}$ (involving α and β) and their $\mathbf{n}$

M1

Legitimately getting given system of equations:

$$3\alpha - 2\beta + 2\lambda = 0$$

$$4\alpha - 2\beta - 5\lambda = -44$$

$$7\alpha - 3\beta + 2\lambda = -11$$

A1

3

- (ii) Solving this 3×3 system in α , β and λ
(possibly just λ given or both α , β)

M1

For $\alpha = -2$, $\beta = 3$ ($\lambda = 6$)

A1

$A = (1, -17, -5)$ and $B = (13, 13, 7)$

Give one A1 for a correct pair (α , A) or (β , B)

A1

SC: allow all 3 marks ft for misreads of the signs
(44 and / or 11 only)

3

(iii) Shortest distance = $\sqrt{12^2 + 30^2 + 12^2}$

or $|\lambda| |\mathbf{n}| = 6\sqrt{2^2 + 5^2 + 2^2}$

Allow also for the shortest distance formula $|(\mathbf{b} - \mathbf{a}) \cdot \hat{\mathbf{n}}|$

M1

$6\sqrt{33}$, $\frac{198}{\sqrt{33}}$, $\sqrt{1188}$
or AWRT 34.5

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CAO

A1

2

[10]

Q6.

(a) $0 = 2 + \lambda \Rightarrow \lambda = -2$

M1

Check $-1 + -2 \times -3 = -1 + 6 = 5$

$-5 - 2 \times 2 = -5 \times -4 = -9$

OE

A1

2

$$(b) \quad \overrightarrow{BC} = \begin{bmatrix} 9 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 5 \\ -9 \end{bmatrix} = \begin{bmatrix} 9 \\ -3 \\ 12 \end{bmatrix}$$

$$\pm (\overrightarrow{OC} - \overrightarrow{OB})$$

M1A1

2

$$(c) \quad (i) \quad \overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD} = \overrightarrow{OA} + 2\overrightarrow{BC}$$

M1

$$\overrightarrow{OD} = \begin{bmatrix} 2 \\ -1 \\ -5 \end{bmatrix} + \begin{bmatrix} 18 \\ -6 \\ 24 \end{bmatrix} = \begin{bmatrix} 20 \\ -7 \\ 19 \end{bmatrix}$$

D is (20, -7, 19)

AG

A1

2

$$(ii) \quad \overrightarrow{PD} = \overrightarrow{OD} - \overrightarrow{OP} =$$

$$\begin{bmatrix} 20 \\ -7 \\ 19 \end{bmatrix} - \begin{bmatrix} 2+p \\ -1-3p \\ -5+2p \end{bmatrix} = \begin{bmatrix} 18-p \\ -6+3p \\ 24-2p \end{bmatrix}$$

Find $\overrightarrow{PD}$ in terms of p

M1

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condone $\overrightarrow{PD} = \overrightarrow{OP} - \overrightarrow{OD}$ here

A1

$$\overrightarrow{PD} \cdot \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = 0$$

B1

$$(18-p) \times 1 + (-6+3p) \times -3 + (24-2p) \times 2 = 0$$

m1

$$p = 6$$

CSO OE working with $\overrightarrow{DP}$

A1

Q7.

(a) (i) $\begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix} = 0$

Equating dot product to zero

M1

$$24p + 6 - 3p + 6 + 2 = 0$$

Solving a linear eqn. in t

M1

$$p = -\frac{2}{3}$$

CAO

A1

3

(ii) $\begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} = m \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix}$

M1

$$m = 2, p = \frac{1}{2}$$

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A1,A1

3

ALT $\begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} \times \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix} = \mathbf{0} \quad \begin{bmatrix} 1-2p \\ 8p-4 \\ 18p-9 \end{bmatrix}$

(M1A1)

$$p = \frac{1}{2}$$

(A1)

(b) (i) $6x - 3y + 2z = 42, 17x + 2y + z = -7$

B1,B1

2

- (ii) eg 2. $\textcircled{2} - \textcircled{1} : 7 [4x + y = -8]$
Eliminating one variable; correct

M1 A1

$$\frac{x+2}{-1} = \frac{y}{4} = \lambda$$

Parametrisation attempt

M1

Substituting back to find 3rd variable

m1

$$z = 9\lambda + 27$$

A1

$$\mathbf{r} = \begin{bmatrix} 0 \\ -8 \\ 9 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -12 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} -2 \\ 0 \\ 27 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix}$$

ft (any pt. on l)

B1

OR method for finding the d. v.

$$\begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix}$$

(M2A1)

Method for finding any pt. on l

(M1A1)

Putting them together as a line eqn.

ft

(B1)

6

- (c) $\mathbf{r} = \mathbf{a} + u \mathbf{d}_1 + v \mathbf{d}_2$
 where $\mathbf{a}$ is an pt. on plane
ft their previous $\mathbf{a}$; or (30, 7, 30)
- $\mathbf{d}_1 =$ any d. v.
ft their previous d. v.

$$\mathbf{d}_2 = \begin{bmatrix} 30 \\ 7 \\ 30 \end{bmatrix} - \text{any } \mathbf{a} \text{ in plane}$$

ft if is clear where it has come from

2

[16]

Q8.

(a) (i)
$$\overrightarrow{AB} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$\pm (\overrightarrow{OA} - \overrightarrow{OB})$

M1

A0 if answer as coordinates

A1

2

(ii)
$$\overrightarrow{OB} \cdot \overrightarrow{AB} = 3 \times 1 + 1 \times 0 + (-2) \times (-1) = 5$$

Evaluate to single value

M1A1

$$\cos \theta = \frac{\overrightarrow{OB} \cdot \overrightarrow{AB}}{|\overrightarrow{OB}| \times |\overrightarrow{AB}|}$$

Use formula for $\cos \theta$ with any 2 vectors and at least one of the corresponding moduli 'correct'

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M1

$$|\overrightarrow{OB}| = \sqrt{14} \quad |\overrightarrow{AB}| = \sqrt{2}$$

$$\cos \theta = \frac{5}{\sqrt{7} \times 2 \sqrt{2}} = \frac{5}{2\sqrt{7}}$$

CSO; AG so need to see intermediate step

$$\text{eg } \frac{5}{\sqrt{7} \times 2 \sqrt{2}} \text{ or } \frac{5}{\sqrt{28}}$$

A1

Alternative

cos rule attempted with cos B

(M1)

cos rule correct with cos B

(A1)

derive correct given form

(A2)

4

(b)
$$\mathbf{r} = \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$\overrightarrow{OC} + \lambda \overrightarrow{AB}$. Allow one slip

M1

ft on $\overrightarrow{AB}$; needs $\mathbf{r}$ or $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

A1F

2

(c)
$$\overrightarrow{OD} \cdot \overrightarrow{AB} = \begin{bmatrix} 6 + \lambda \\ 2 \\ -4 - \lambda \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

M1

$$6 + \lambda + 4 + \lambda = 0$$

m1

$\lambda = -5$ ©2025 Exam Papers Practice. All rights reserved.

ft on equation of line

A1F

D is (1, 2, 1)

CAO

A1

Alternative

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = a - c = 0$$

Let D be (a, b, c)

Scalar product evaluated and equated to 0

(M1)

$$a = 6 + \lambda, \quad b = 2, \quad c = -4 - \lambda$$

Use equation of line

(m1)(A1)

$$a + c = 2$$

$$a = 1 \quad b = 2 \quad c = 1$$

(A1)

4

[12]

Q9.

(a) $\mathbf{n} = (3\mathbf{i} + \mathbf{j} + 2\mathbf{k}) \times (4\mathbf{i} - \mathbf{j} + \mathbf{k})$

M1

$$= 3\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$$

cao

A1

$$d = (2\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot (\text{their } \mathbf{n}) = 4$$

ft

M1A1

4

(b) $\begin{bmatrix} 7 + 10t \\ 1 + t \\ 4 + 5t \end{bmatrix}$ substd. into their plane eqn.
(In at least the LHS of it)

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$$21 + 30t + 5 + 5t - 28 - 35t = 4$$

Linear "eqn." in t created (LHS)

dM1

Since $-2 \neq 4$, no intersection

*Explained or stated. N.B. can ft
other d's (except -2) but if n is wrong
also the t won't vanish, so no ft then*

A1

Line parallel to plane

May be independently asserted

B1

OR

$$\begin{bmatrix} 3 \\ 5 \\ -7 \end{bmatrix} \cdot \begin{bmatrix} 10 \\ 1 \\ 5 \end{bmatrix} = 0$$

(M1)(A1)

For showing line not in plane

(B1)

Line perp^r. to nml. $\Rightarrow$ line // to plane

(B1)

OR

$$\begin{bmatrix} 7 + 10t \\ 1 + t \\ 4 + 5t \end{bmatrix} \text{ equated to } \begin{bmatrix} 2 + 3\lambda + 4\mu \\ 1 + \lambda - \mu \\ 4 + 2\lambda + \mu \end{bmatrix}$$

Incl. starting to do something

(M1)

Eliminating λ, μ to get linear eqn. in t

(dM1)

Since $-2 \neq 4$, no intersection

Explained or stated

(A1)

Line parallel to plane

May be independently asserted

(B1)

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4

[8]

Q10.

(a) (i)
$$\overrightarrow{AB} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ -3 \end{bmatrix}$$

M1 for $\pm (\overrightarrow{OA} - \overrightarrow{OB})$

M1A1

2

(ii)
$$(\mathbf{r} =) \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -4 \\ -3 \end{bmatrix}$$

ft on $\overrightarrow{AB}$; OE

B1F

1

(b) (i)
$$\begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 5 \end{bmatrix}$$

μ found and verified or statement $\mu = -3$ satisfies all components

M1

$$1 + \mu = -2 \quad \mu = -3$$

$$-1 - 2\mu = 5 \quad \mu = -3$$

$\mu = -3$ alone B1

A1

ALTERNATIVE METHOD

$$\mu \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} \text{ which is satisfied by } \mu = -3$$

2

(ii)
$$\overrightarrow{PQ} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ -3 \end{pmatrix} - \begin{pmatrix} -2 \\ -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 + 2\lambda \\ 8 - 4\lambda \\ -4 - 3\lambda \end{pmatrix}$$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$ with $\overrightarrow{OQ}$ in parametric form in terms of λ (can be inferred later)

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M1

$$\text{or } \begin{bmatrix} 6 + 2\lambda \\ 4 - 4\lambda \\ -7 - 3\lambda \end{bmatrix}$$

A1

$$\begin{bmatrix} 4 + 2\lambda \\ 8 - 4\lambda \\ -4 - 3\lambda \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

$$\overrightarrow{PQ} \cdot \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \text{ with } \overrightarrow{PQ} \text{ in terms of } \lambda$$

(can be inferred later)

M1

$$(4 + 2\lambda) + (-2)(-4 - 3\lambda) = 0$$

linear expression in λ equated to 0

m1

$$\lambda = -1.5$$

ft on sign/arithmetic error in $\overline{PQ}$ or equation

A1F

Q is (-1, 11, 5.5)

CAO

A1

6

[11]

Q11.

(a) $\begin{bmatrix} 3 \\ -3 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} = 3 - 6 + 3 = 0$

attempt at sp, 3 terms, added

M1

$= 0 \Rightarrow$ perpendicular

$= 0 \Rightarrow$ perpendicular seen (or $\cos \theta = 0 \Rightarrow \theta = 90^\circ$)

$\begin{matrix} 3 \\ -6 \\ 3 \\ 0 \end{matrix}$ $\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} = 0$

Allow $\frac{3}{0}$ but not $\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$

A1

2

(b) $8 + 3\lambda = -4 + \mu$

$$6 - 3\lambda = 2\mu$$

$$-9 - \lambda = 11 - 3\mu$$

$$\lambda = -2, \mu = 6$$

set up any two equations

M1

solve for λ and μ

m1 A1

verify third equation

substitute λ, μ in third equation

m1

intersect at (2, 12, -7)

CAO

A1

5

Alt (for last two marks)

substitute λ into l_1 and μ into l_2

(m1)

intersect at (2, 12, -7), condone $\begin{pmatrix} 2 \\ 12 \\ -7 \end{pmatrix}$

(2, 12, -7) found from both lines

Note: working for (b) done in (a): award marks in (b)

(A1)

(c) $\vec{AP} = \begin{pmatrix} 6 \\ 12 \\ -18 \end{pmatrix}$

$$\vec{AP} = \pm \left\{ \text{their } \vec{OP} - \begin{pmatrix} -4 \\ 0 \\ 11 \end{pmatrix} \right\}$$

M1

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$$AP^2 = 504$$

ft on p

A1F

$$AB^2 = 2AP^2$$

Calculate AB^2

M1

$$AB = 12\sqrt{7}$$

OE accept 31.7 or better

A1

4

[11]

Q12.

- (a) $\lambda = -4$ gives $P(-29, 42, -19)$ on l
Correct value of λ

B1

1

- (b) $\cos^{-1} \frac{1}{9}$ or 83.6° (or 84°) or 1.46 rads.
ft from 3rd d.c. or by any other method
(e.g. scalar product)
N.B. Mark lost if 6.4° is then offered as the answer

B1 ✓

1

- (c) (i) $\mathbf{n} = 3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

B1

1

- (ii) Use of $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$
Must be direction vector of l and their $\mathbf{n}$

M1

Nr. = 45 Dr. = $\sqrt{50}.9$
ft the "9" if necessary from (b)(i)

A1 A1

$\theta = 45^\circ$

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A1

4

- (d) Substⁿ. $\begin{pmatrix} 3+8\lambda \\ 26-4\lambda \\ \lambda-15 \end{pmatrix}$ in $3x - 4y + 5z = 100$
 $3(3 + 8\lambda) - 4(26 - 4\lambda) + 5(\lambda - 15) = 100$

M1

Solving a linear eqn. in λ

dM1

$\lambda = 6$

CAO

A1

$$\Rightarrow Q = (51, 2, -9)$$

ft their λ in l

B1 ✓

4

(e) $PQ = \sqrt{80^2 + 40^2 + 10^2} = 90$

ft

B1

$$\text{Sh. Dist.} = 90 \sin 45^\circ = 45\sqrt{2} \text{ or } 63.6(4\dots)$$

M1

ft

A1 ✓

3

Or $p + m n$ subst^d. into $\Pi \Rightarrow m = 9$

(M1)

$$\Rightarrow R = (-2, 6, 26)$$

R = foot of perp^r. from P to Π

(A1)

$$PR = \sqrt{27^2 + 36^2 + 45^2} = 45\sqrt{2}$$

ft

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(B1 ✓)

(3)

[14]

Q13.

(a) (i) $\mathbf{n} = (3\mathbf{j} + \mathbf{k}) \times (4\mathbf{i} - \mathbf{j}) = \mathbf{i} + 4\mathbf{j} - 12\mathbf{k}$

M1A1

2

Alternative:

$$\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \quad \text{and/or} \quad \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

(M1)

$$\Rightarrow c = -3b, b = 4a \text{ i.e. } a(\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$$

(A1)

(2)

(ii) $d = (2\mathbf{i} + 5\mathbf{j} + \mathbf{k}) \cdot (\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}) = 10$

Their n

M1A1 ✓

2

(b) Either substituting.

$$r = \begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix} \text{ into } \left(r = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

M1

And showing result is 0

A1

2

Or

Explaining p.v. and d.v. of line

Give the M1 for either

M1A1

2

(c) Substⁿ $r = \begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix}$ into their plane eqn.

M1

Solving a linear eqn. in t

$$2 + 3t + 8 - 60 + 12t = 10$$

dM1

$$t = 4$$

A1 ✓

Pt. of intersection at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

CAO

B1

4

First alternative (c)

Uses (b)'s given form for L and the original form for Π

Setting $\begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix} = \begin{bmatrix} 2+4\mu \\ 5+3\lambda-\mu \\ 1+\lambda \end{bmatrix}$

Must make an attempt to solve

(M1)

$$\mu = \frac{3}{4}t, \mu = 3 + 3\lambda, \lambda = 4 - t$$

(A1)

Eliminating λ and μ (e.g.) $\Rightarrow t = 4$

(dM1)

Pt. of i/sctn. at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

CAO

(A1)

4

Second alternative (c)

Uses other given form for L

$$\left(\begin{bmatrix} 2+4\mu \\ 5+3\lambda-\mu \\ 1+\lambda \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \text{ gives}$$

$$\begin{bmatrix} 3+4\mu \\ 3+3\lambda-\mu \\ \lambda-5 \end{bmatrix} \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -3-3\lambda+\mu \\ 3\lambda+4\mu-12 \\ 3\mu-9\lambda-9 \end{bmatrix}$$

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(M1 A1)

Equating to $\mathbf{0}$ and solving

$$\Rightarrow \lambda = 0, \mu = 3$$

(M1)

Pt. of intersection at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

CAO

(A1)

4

[10]

Q14.

(a) (i) $\vec{OC} = 2 \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ -2 \end{bmatrix}$
(Penalise coordinates once only)

B1

1

(ii) $\vec{AB} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$
 $\vec{OA} - \vec{OB}$ or $\vec{OB} - \vec{OA}$ or 2/3 correct components.

M1

A0 for line AB

A1

(b) (i) $AC^2 = (6-2)^2 + (4-4)^2 + (-1-2)^2 = 25$
|Components of AC|

M1

$AC = 5$
AG

A1

2

(ii) $\vec{AB} \cdot \vec{AC} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 0 \\ -3 \end{bmatrix} = 4 + 6 = 10$
Clear attempt to use $\vec{AB}$ and $\vec{AC}$

M1

ft $\vec{AB}$ from a(ii) and/or $\vec{AC}$ from b(i)

A1F

$3 \times 5 \times \cos \theta = 10$

Use of $|\mathbf{a}| |\mathbf{b}| \cos \theta = \mathbf{a} \cdot \mathbf{b}$

With one correct $||$ and $\mathbf{a} \cdot \mathbf{b}$ evaluated

M1

$\theta = 48.189 \approx 48^\circ$

CAO (AWRT)

A1

Alternative: use of cos rule
Find 3rd side + use cos rule

(M2)

ft on previously found vectors

(A1F)

CAO (AWRT)

(A1)

(c) $\overrightarrow{BP} = \begin{bmatrix} \alpha - 3 \\ \beta - 2 \\ \gamma - 1 \end{bmatrix}$

B1

$$\begin{bmatrix} 4 \\ 0 \\ -3 \end{bmatrix} \cdot \overrightarrow{BP} = 0$$

Their $\overrightarrow{BP}$

M1

$$4\alpha - 3\gamma - 15 = 0$$

AG convincingly obtained

A1

EXAM PAPERS PRACTICE [12]

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Q15.

(a) $|\overrightarrow{AB}| = \sqrt{(2-1)^2 + (-1-4)^2 + (3-2)^2}$

OE

M1

$$= \sqrt{27} = 3\sqrt{3}$$

AG; convincingly obtained

A1

2

(b) $\begin{bmatrix} 1 \\ -5 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 1 + 5 + 1$

Their $\vec{AB} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

M1

Their scalar product = $\begin{vmatrix} 1 \\ -1 \\ 1 \end{vmatrix} 3\sqrt{3} \cos \theta$

M1

$7 = 3 \sqrt{3} \sqrt{3} \cos \theta \quad \cos \theta = \frac{7}{9}$
AG; convincingly obtained

A1

3

(c) (i) $\vec{OP} = \begin{bmatrix} 2+p \\ -1-p \\ 3+p \end{bmatrix} \quad \vec{AP} = \begin{bmatrix} 1+p \\ -5-p \\ 1+p \end{bmatrix}$
Finding $\vec{AP}$

M1A1

$\vec{AP} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = (1+p) - (-5-p) + (1+p)$

m1

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$= 7 + 3p$

AG

SC working with λ instead of p giving

$7 + 3\lambda \quad \frac{3}{4}$

A1

4

(ii) $7 + 3p = 0$

M1

$p = -\frac{7}{3} \quad P \text{ is } \left(-\frac{1}{3}, \frac{4}{3}, \frac{2}{3} \right)$

Allow column vectors and decimals

A1A1

3

[12]

Examiner reports

Q2.

Although most students scored well on parts (a) and (b), many made little further progress and generally the question was not done well. However, about 15% of the entry did score 10 or more of the marks available. There were few fully correct solutions to part (d). Notation was often poor and many students did not make it clear which vectors they were working with.

- (a) Most students answered this correctly. Many found $\lambda = -1$ explicitly or wrote a correct vector expression in which it was implied. Most demonstrated that this value satisfied the three components although a few students failed to carry out this step. The other rarely seen error was to find $\lambda = +1$. It was good to see some students writing a clear conclusion to their solution, although this was not required.
- (b) Most students knew they had to find the vector $\overrightarrow{AB}$ and apart from the occasional sign error they did so successfully. However, some thought this was the equation of the line. Many others had the correct form of the line but omitted $\mathbf{r} =$, or wrote $line =$ or something similarly inappropriate for the vector equation of a line.
- (c) Although most students knew they were to use the equation of the line and equate a scalar product with vector $\overrightarrow{AB}$ or $\overrightarrow{AD}$ to zero, many used $\overrightarrow{OD}$ rather than $\overrightarrow{CD}$ and so could gain no further marks. Those who tried to find vector $\overrightarrow{OD}$ often made a sign or coefficient error, so could only score the method marks in finding the value of μ and the coordinates of D . Some got as far as $17 + 17\mu = 0$ and concluded $\mu = 1$.
- (d) Many students did not attempt this part of the question. Success here would have been more likely if a correct diagram had been used, but most of those who attempted this part of the question did not draw one and many had the right angle in the wrong place, at C instead of at D . Many then used the sides AC and AD of the triangle ACD to attempt to find the area whereas the base and height are AD and CD respectively. Some students who chose this route, especially if they had an error in the coordinates of D , continued working with very unfriendly fractions and attempts at finding moduli before abandoning the attempt for no credit. The key to a successful solution lies in realising that CD is the height of both triangles ACD and ACE and as such there is no need to actually find its length, although students found that CD cancels if they did this correctly. Some students did give a concise and correct solution for both the possible positions of point E .

Q3.

This topic of equations of lines and planes is not well understood by students. Surprisingly, part (a) was not always answered correctly with arguments concerning the scalar product often wrongly used. Part (b) was most often successful when two points on the line were used or if the scalar product of the normal to the plane and direction vector of the line was set equal to zero and solved to find p . Part (c)(i) proved very discriminating with students using both scalar and vector products in tandem but not using the correct trigonometric ratio and confusing sine and cosine.

Q4.

In part (a), most students set up the simultaneous equations required correctly, with many of these spotting the solution $\mu = -1$ immediately and thus finding the value of λ easily. Other students used conventional simultaneous equations methods, which took rather longer but could still be done successfully. Most then went on to use the third equation to show that $q = 4$, with relatively few assuming $q = 4$ and showing that it works. Most found the coordinates of the point P correctly, with the occasional coefficient error, although most gave their solution as a column vector rather than coordinates, which was condoned this time.

In part (b), most students knew that they were to show a scalar product came to zero, and most correctly choose the direction vectors and demonstrated this was the case. There was no credit for just stating the result, and students who failed to state the conclusion to their scalar product $= 0$ were also penalised.

Many students misunderstood the notation in part (c)(i), not realising they were being asked to find the square of the length AP . Some students squared their coordinates of point P , before subtracting those of point A , and leaving it as their answer, whereas some multiplied the corresponding coordinates. Some students showed some understanding in getting an answer of $(16, 0, 4)$ but left the answer as a column vector and not a length squared, so got no credit. Some students who had misunderstood part (c)(i) found the length of AP correctly, when attempting part (c)(ii).

In order to make any progress with part (c)(ii), students needed to express the vector $\overrightarrow{BP}$ or $\overrightarrow{AB}$ in terms of a parameter, find the length of the vector and use the properties of the isosceles triangle to set up an equation. There were some good attempts at this stage although rather fewer went on to complete fully successfully. Errors were made in expanding expressions such as $(a + bp)^2$ or in the collection of terms, although many did get to, and attempted to solve, a quadratic equation as required and gained credit for using a correct method. Some students attempted to use the scalar product $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$ but this just reduces to $0 = 0$ if done correctly. There is an alternative method avoiding the use of a parameter, based on equating vector $\overrightarrow{BP}$ to a multiple of the direction vector of line l_2 , but this was rarely seen.

Q5.

Part (a) was a routine and straightforward demand, as was part (b)(i), and most students gained these five marks. Solving the given system of equations was found to be less straightforward, although some students were able to use their calculators to reach an answer with relative ease.

As commented on in previous reports, students are reluctant to draw a quick, simple diagram that could direct their thoughts to what is required. Despite the fact that the whole thrust of the question had led to part (b)(iii), surprisingly few students really seemed to know what they were finding here, even when they had correctly completed all the preceding parts of the question.

Q6.

Most candidates answered part (a) well using a variety of methods. Virtually all gave the value of λ as -2 , either explicitly or implied by its use, and then completed a full justification. A common error was to verify only one of the two other coordinates, or use poor notation that was not convincing.

In part (b), apart from the occasional arithmetic error, all candidates found the vector correctly.

In part (c)(i), most candidates went on to use the result from part (b) correctly to find the coordinates of D , although most left the result as a vector rather than in coordinates. Many candidates, though, used poor notation with, for example, D representing the vector $\vec{OD}$.

The key to success in part (c)(ii) was in noting that the scalar product between $\vec{PD}$ and the direction vector of the given line was zero. Although some clear and correct answers were seen, this part defeated many candidates, although some might have gained some credit had they been clear which vectors they were trying to work with. Many candidates did not attempt to express the vector $\vec{PD}$ in terms of λ and so could make little progress. Similarly, those candidates who let the point P have coordinates (x, y, z) were unsuccessful often abandoning a complex looking algebraic expression. Those who made some progress tried to take the scalar product with a point on the line rather than its direction, and produced largely insoluble equations, which were abandoned, or a value of p just appeared. Some candidates who clearly understood what they were doing made sign or coefficient errors when finding $\vec{PD}$; a common wrong answer was $p = -6$.

Q7.

The structure of this question proved very helpful to candidates, and those who were reasonably careful could generally score at least 13 of this question's 16 marks without any difficulty. The main hurdle arose where many candidates failed to realise in part (b) that they had already worked out the eigenvalues of $\mathbf{M}$ in part (a), and thus they went ahead and started all over again in an effort to find, and then solve, a cubic characteristic equation. This was all a bit of a waste of time and frequently not as successful as their former effort.

Q8.

Most candidates used an appropriate method in part (a)(i) to find the requested vector, with some making a sign error, particularly with the z component, where $-1 - 2 = -3$ was common. Some candidates expressed the vector as if it were coordinates and were penalised. For part (a)(ii), although many candidates did choose the correct vectors for the scalar product and had them clearly identified, some worked with the position vectors $\vec{OA}$ and $\vec{OB}$. Of these, some proceeded to get their own incorrect answer, whilst others were determined to get to the given result for $\cos \theta$, without apparently considering that they had made an error.

Some candidates had the negative of one of the vectors they should have been working with and just dropped the resulting minus sign. Some candidates, who were otherwise correct, were denied a mark for not showing how $\sqrt{14}\sqrt{2}$ became $2\sqrt{7}$. For part (b), most candidates knew the $\mathbf{a} + \lambda\mathbf{b}$ form for the vector equation of a line, although few used the correct notation of $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and just called it $\mathbf{l} = \mathbf{a} + \lambda\mathbf{b}$, which was penalised. Some candidates had the position vector $\mathbf{a}$ and the direction vector $\mathbf{b}$ reversed.

There were relatively few successful attempts at part (c). Many candidates knew they had to equate a scalar product to zero, but attempted $\vec{OD} \cdot \vec{OC}$ or used the parallel vectors $\vec{AB}$ and $\vec{CD}$. A diagram would probably have helped to identify the correct vectors to use. Some did use $\vec{CD}$ in place of $\vec{AB}$ which was correct, but candidates were more prone to error in solving the resulting quadratic equation. However, some well presented

fully correct solutions were seen.

Q9.

This was the first question on the paper which required some explanation by candidates, to justify their working in some way or to explain their results. Although responses were a *lot* better than they generally have been in the past, there is still much scope for improvement. In particular, candidates need to realise that, when the answer is given in the question, they need to be a little more diligent in its justification. This especially applies to the better candidates, who can frequently “see” things as obvious or work them out “in their heads”. Sloppy presentation of solutions can often lose these candidates marks. In part (b), the question tells them that the line and plane do not intersect. Even amongst those who substituted the given line equation into the correct plane equation, and found a contradiction of the “ $-2 = 4$ ” variety (see the mark scheme), very few offered a satisfactory explanation of the non-intersection. A small number of candidates took the slightly less obvious approach of showing that the line was perpendicular to the plane’s normal; however, this did not, on its own, establish that the line didn’t lie *in* the plane, and an extra step was needed in order to gain all four marks here.

Q10.

This question also separated clearly into two parts, with most candidates scoring well in part (a) and part (b)(i) but few making substantial progress in part (b)(ii).

Part (a)(i) was usually correct but with the occasional sign error or the position vectors the wrong way round. In part (a)(ii), some candidates chose to give one of the position vectors as their direction vector.

There were several approaches to part (b)(i), most of which were acceptable, the important thing here being to give a convincing argument that the given point does lie on the line. This was best done by finding the value of μ from one component equation and showing that it satisfies the others.

Candidates whose working was unclear, with $\mu = -3$ appearing somewhere within it, usually only scored 1 mark. Candidates should always state which vectors they are

working with in terms of letters, such as $\overrightarrow{AB}$, rather than giving numerical values which might not be recognisable. This was particularly the case in part (b)(ii), where it often was not clear what candidates were trying to do as they had not identified the vectors with which they were working. However, it was apparent that, of those who did make a

recognisable start, some tried to work with vector $\overrightarrow{OQ}$ instead of $\overrightarrow{PQ}$ and then often went on to take the scalar product with the wrong vector. Some candidates tried to work with a

vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, again often without saying what it represented, and, although a few eventually expressed x , y and z in terms of λ , by then many had made a mistake and it became difficult to follow how they had got to their incorrect result. The use of the parametric equation of the line to solve a problem of this nature appeared not to be well known.

Q11.

In part (a), most candidates were successful, both showing that they knew the condition for two lines to be perpendicular and demonstrating that the scalar product was zero. A

few failed to state the conclusion after finding a zero scalar product. Quite a number of candidates started their solution to part (a) as if they were answering part (b), with some stating the conclusion that the lines intersect and so are perpendicular. Although this gained no credit for part (a), it was credited in part (b).

In part (b), most candidates knew they were to set up simultaneous equation and solve them which most did successfully. Both elimination and substitution techniques were seen, the latter usually requiring rather more work. The common errors were in the signs. Some candidates failed to verify that an independent third equation was also satisfied by their solutions or that their intersection point indeed lay on both lines. Some, who attempted verification and found that it did not work, did not then check their own working for an error; they simply concluded that the lines did not intersect, with some adamantly stating that the question was wrong.

Fully correct solutions to part (c) of the question were rare. Many candidates got no further than finding the vector $\vec{AP}$ from their previous result, then taking the question to mean $\vec{AP} = \vec{BP}$ rather than that the lengths of these two vectors were the same. Those candidates who did realise that the question was about lengths of vectors and drew a diagram were generally successful. Others were determined that they could find the point B by using an arbitrary point on the line l , often assuming $\lambda = 0$. Many concluded that A and B were the same point and so the distance between them was zero.

Q12.

Parts (a) to (d) of this question were usually very well done, and candidates were able to score a lot of the marks. The big surprise came in part (e) when so few arrived at the correct final answer. In almost all cases, a simple diagram would have helped enormously, and they would then have seen that a simple bit of basic trigonometry would have done the trick. In most cases, candidates seemed to be working with randomly-selected vectors from the first few parts of the question and trying to insert them into some (often half-) remembered vector formula.

Q13.

This was one of the questions which candidates with a weak grasp of the underlying ideas behind topics found difficult. Part (a)(i) can be done either by a single vector product or by two scalar products. A large number of candidates then used one of the plane's two direction vectors, rather than the position vector of the point $(2, 5, 1)$ which lies in the plane Π , to find the scalar constant d required. This is one of several fundamental errors made by candidates.

Part (b) was a straightforward test of the alternate form of a line equation, yet it was frequently answered in unconvincing ways. The instruction to "verify" – that is, to substitute the given expression for $\mathbf{r}$ into the given equation for L – was clearly not understood by many. The perfectly allowable alternative approach is to explain the role of the line's direction vector and show that the point $(2, 2, 5)$ is a point on the line. Many candidates managed one or the other but there was often a confusing amount of vector algebra and arithmetic presented with no apparent purpose and absolutely no explanation. Such vague approaches usually gained one of the two marks, but not both.

Part (c) was generally done quite well, as there are at least three approaches, at least one of which does not require successful progress in any previous part of the question.

Q14.

Most candidates gained some marks on the vectors question, particularly in parts (a) and (b)(i). Some candidates persist in using coordinate notation instead of components, and this was penalised.

In part (a) although the vector $\overrightarrow{OC}$ could just be written down, some candidates subtracted the zero vector before doubling the result. Virtually all candidates knew how to find the vector $\vec{AB}$, although sign and arithmetic errors were fairly common.

Part (b)(i) Most candidates did this successfully by first finding the vector $\overrightarrow{AC}$ and then calculating its modulus. Those candidates who did not show why they were squaring 3 and 4 to get the result, were penalised. In part (b)(ii), most candidates showed that they can calculate an angle between two vectors, but rather fewer showed they could choose the correct two vectors with the correct directions. The angle was given in the question as BAC and so candidates were expected to work with $\vec{AB}$ and $\overrightarrow{AC}$ or $\overrightarrow{BA}$ and $\overrightarrow{CA}$. Many candidates had a direction incorrect on one of these vectors, whereas many more used position vectors, or vectors where it just was not clear where they had got them from. Some candidates calculated the lengths of the sides of the triangle and used the cosine rule, often successfully.

Part (c) Answers ranged here from a well set out high quality demonstration of the result, to considerable illicit manipulation of vectors to get the given result. Most candidates knew a zero scalar product was involved, but many were not at all clear in what way. Here, and also in part (b), some candidates gave their evaluated scalar product in the form of a column vector, rather than as a scalar which is incorrect and was penalised. Some just worked with the vector $\overrightarrow{OP}$ but still managed to “derive” the result.

Q15.

Part (a) This was usually done well with most candidates obtaining the components 1, 5, 1 and showing the connection to $\sqrt{27}$.

Part (b) Here there was an unexpected lack of knowledge of the scalar product. Although $|a||b|\cos\theta = a \cdot b$ was often quoted, many candidates left their scalar product in a column and few could convincingly show where $7/9$ came from.

Part (c)(i) This was seen more so here, with scalar products written as columns and the given answer just written down. Candidates were expected to find $\vec{AP}$ using $\overrightarrow{OP} - \overrightarrow{OA}$ and relatively few did it convincingly.

Part (c)(ii) Most candidates understood that they needed to put $7 + 3p = 0$ and went on to solve for p and to find the coordinates correctly, although some made arithmetical errors.